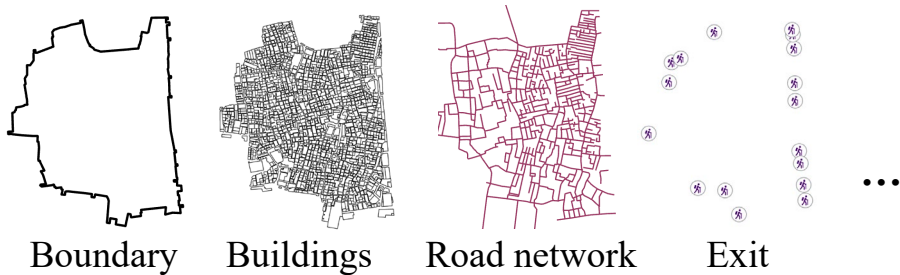
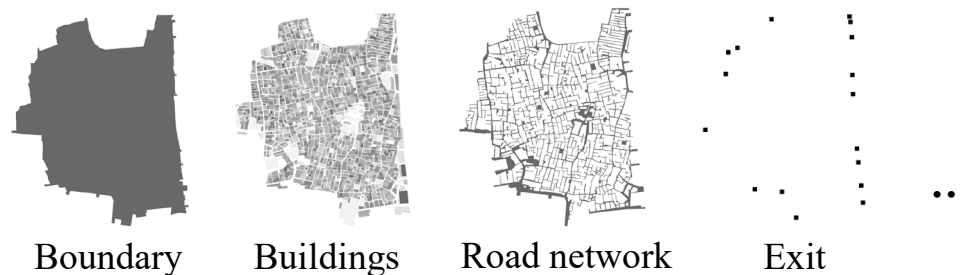


Vector GIS Inputs



Multi-layer Raster

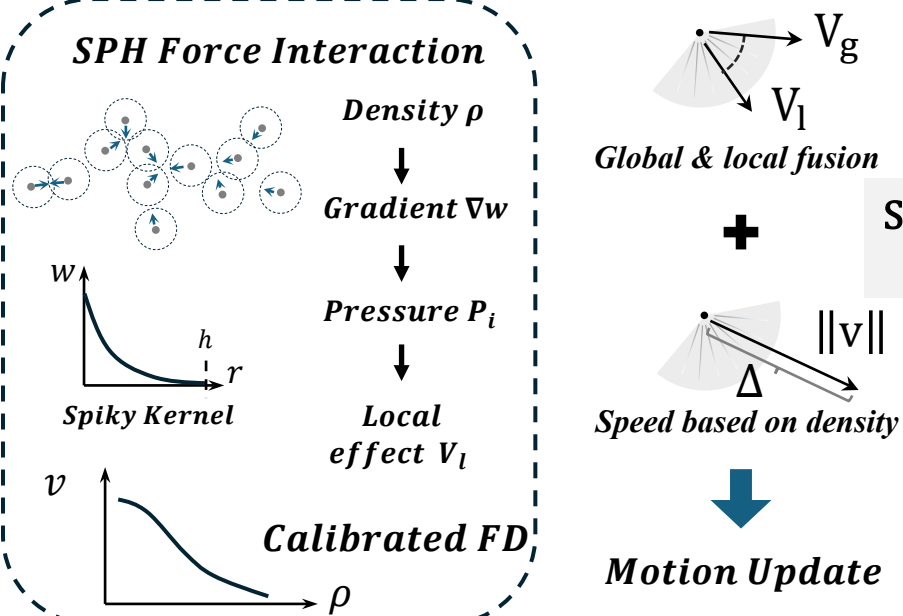


I. Computational Environment Construction



Global guidance $d_{nav}(x)$

II. Crowd Evacuation Simulation



Layout update

III. Targeted Micro-Regeneration

Cost-effective road-widening plan

Pareto front

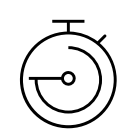
Selected compromise

Simulation-in-the-loop NSGA-II

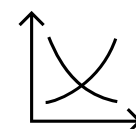
Minimize two objectives

Map update

SPH evaluation



$F_{evac}(x)$

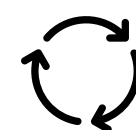


$F_{cost}(x)$

$M(x)$

$\delta_k \in 0,2,4m$

Widening



Simulation Rollout

First Simulation for Bottleneck identification

